

VPN

- A VPN, as the name implies, is a means of creating a private network across an untrusted network such as the internet.
- A virtual private network connection allows you to access the internet through a remote server, hiding your actual location and browser history, and encrypting your data
- VPNs can be used for a number of different purposes such as:
  - to securely connect isolated local area networks (LANs) across the internet
  - to allow mobile users remote access to a corporate network using the internet
  - to control access within an intranet environment.
- VPN service providers like SurfShark, Express VPN and NordVPN
- Free VPNS : Hide.me, PRIVADOVPN or ProtonVPN unfortunately these free VPNs have speed or data limits

VPN

- There are at least three situations in which a VPN is the ideal solution:
  - One or more workers need to occasionally work from home (as in the scenario described above). In this situation, a **VPN connection establishes a connection between the home computer and the office network.**
  - One or more mobile users — who may not ever actually show up at the office — need to connect to the work network from mobile computers, often from locations like hotel rooms, clients' offices, airports, or coffee shops.
    - This type of VPN configuration is similar to the home user's configuration, except that the exact location of the remote user's computer is not fixed.
  - Your company has offices in two or more locations, each with its own local area network, and you want to connect the locations so that users on either network can access each other's network resources.
    - In this situation, the VPN doesn't connect a single user with a remote network; instead, it connects two remote networks to each other.

VPN

- VPNs provide the following:
  - **Confidentiality**: preventing anyone from reading your data. This is implemented with encryption.
  - **Authentication**: verifying that the router/firewall or remote user that is sending VPN traffic is a legitimate device or router.
  - **Integrity**: verifying that the VPN packet wasn't changed somehow during transit.
  - **Anti-replay**: preventing someone from capturing traffic and resending it, trying to appear as a legitimate device/user.

The benefits of a VPN connection

1. Anonymity and privacy:
   - **Anonymity on the internet:** A VPN connection hides your real IP address and location. With a VPN connection, you use the IP address that belongs to the VPN server you're connected to. This way your own IP address stays hidden.
   - **Downloading and uploading anonymously**: Downloading certain Torrents is illegal in some countries and more than ever before downloaders are tracked down and sometimes even prosecuted.
   - **Prevent companies from building a file on you**: Advertising networks such as Facebook, Google, and Twitter are constantly collecting information about you through your online traffic. With this information, they can show you tailored ads but more importantly, they are free to sell this information to a third party. By encrypting your data using a VPN these networks will have a harder time collecting information on you. They will also have less influence on what you see online.

The benefits of a VPN connection

2. **Safety on the web:**
   - **Protection against hackers and governments**: Without a secure VPN connection, it is much easier for others to intercept, view, and steal the data that you, often unknowingly, spread when you browse the web.
   - **Secure internet access on public Wi-Fi networks**: Using a public Wi-Fi network can be very risky. Other users on the same network can easily tap into your data and personal information.
   - Since you don't want others to have access to, for instance, your email login, images/files or credit card information it might be wise to use a VPN connection. The VPN encrypts all of your internet traffic. A hacker will only see encrypted matter and won't be able to see or use your personal information.

The benefits of a VPN connection

3. **Remote access and bypassing internet censorship and geo-blocks**
   - Bypassing censorship and geographical restrictions: For a normal internet user, there can be many restrictions as to what they can and cannot access online.
   - For instance, you cannot watch all the content available on BBC iPlayer if you are not accessing the internet via a British IP address. Also, some streaming services like Netflix offer different content in different geographical zones
   - **Access to your company's network**:
     - More and more companies are giving people the possibility of working from home, or abroad for instance.
     - Some people connect to the internet via a VPN to access the company network at home. This enables people to efficiently work from home.

VPN Clients and Servers

- A VPN connection requires a VPN Server and a VPN Client — the server is the gatekeeper at one end of the tunnel, the client at the other
- The computer initiating the VPN connection is referred to as the VPN client.
- The computer responding to the initiation of the VPN connection is referred to as the *VPN server*.
- The remote computer is the VPN client that makes aremote access connection to the VPN server

VPN Clients and Servers

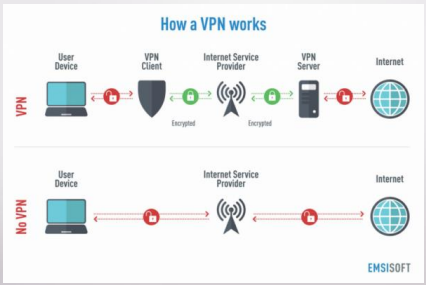


How does a VPN work?

- A VPN hides your IP address by letting the network redirect it through a specially configured remote server run by a VPN host.
- This means that if you surf online with a VPN, the VPN server becomes the source of your data.
- This means your Internet Service Provider (ISP) and other third parties cannot see which websites you visit or what data you send and receive online.
- A VPN works like a filter that turns all your data into "gibberish".
- Even if someone were to get their hands on your data, it would be useless

How Does a VPN Work





How Does a VPN Work

1. With a couple of clicks, you select your preferred security settings and set up a safe connection with your desired VPN server. Connecting to a VPN is generally simple. When the connection has been established, the following will happen to your data:
2. The VPN software on your computer encrypts your data traffic and sends it (via your Internet Service Provider) to the VPN server through a secure connection.
3. The encrypted data from your computer is decrypted by the VPN server.
4. The VPN server will send your data on to the internet and receive a reply, which is meant for you, the user.
5. The traffic is then encrypted again by the VPN-server and is sent back to you.
6. The VPN-software on your device will decrypt the data so you can actually understand and use it.

VPN Protocols

- VPN protocols determine exactly how data is routed through a connection.
- These protocols have different specifications based on the benefits and desired circumstances; for example, some VPN protocols prioritize data throughput speed while others focus on masking or encrypting data packets for privacy and security.
- These protocols ensure that your data remains safe and can't be read by others.
- VPN protocols are similar to a set of instructions.
- VPN providers use these protocols to make sure that users are able to connect securely to a virtual private network.
- VPN Protocols: PPTP, Open VPN,IPSEC/L2TP, IKEv2/Ipsec, SSTP

VPN Protocols: **PPTP**

- **PPTP** (Point to Point Tunneling Protocol) is one of the first widely used protocols.
- Point-to-Point Tunneling Protocol is one of the oldest VPN protocols in existence. Developed in the mid-90s by Microsoft, PPTP was integrated into Windows 95 and specifically designed for dial-up connections.
- This protocol contains a few (potential) leaks. For this reason, its use is only advisable if speed is more important to you than safety.
- This could be the case if you want to bypass restrictions set up by streaming services
- It lacks many of the security features found in other modern protocols it can deliver the best connection speeds for users who may not need heavy encryption.
- But while PPTP is still used in certain applications, most providers have since upgraded to faster more reliable protocols.

VPN Protocols: **OpenVPN**

- OpenVPN is one of the most widely used VPN protocols.
- It's an open-source protocol that uses encryption based on the OpenSSL and SSLv3/TLSv1 protocols.
- Most VPN services support OpenVPN and it's available for a lot of different platforms (e.g. Windows, Android, Linux, routers).
- Most consider OpenVPN to be the best choice. The only downside is, that it doesn't work on Mac OS and iOS, so you need to use a different protocol for that (such as IKEv2/IPsec).

VPN Protocols: IPSEC/L2TP

- Layer 2 Tunnel Protocol(L2TP) is a replacement of the PPTP VPN protocol.
- This protocol does not provide any encryption or privacy out-of-the-box and is frequently paired with security protocol Ipsec, for the encryption of data with L2TP for establishing a secure connection
- Once implemented, L2TP/IPsec is extremely secure and has no known vulnerabilities.
- Most operating systems include IPsec/L2TP, which is a good choice when OpenVPN isn't available

VPN Protocols: **IKEv2/IPsec**

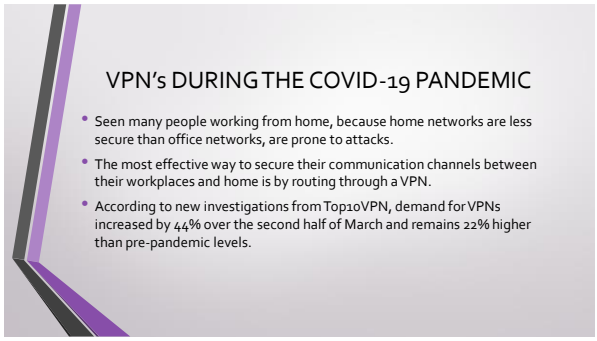
- **IKEv2/IPsec:** IKEv2 is a protocol based on IPSec.
- This protocol can quickly connect to and switch between networks.
- This protocol is very good at re-establishing the link after temporary connection loss and excels at switching connections across network types (from WiFi to cellular, for example).
- This makes it an ideal choice for smartphones because these devices tend to switch between Wi-Fi networks and public Wi-Fi networks regularly.
- IKEv2 is faster than OpenVPN. Nevertheless, OpenVPN is seen as the better protocol.

VPN Protocols: SSTP

- Secure Socket Tunneling Protocol is popular due to its full integration with every Microsoft operating system since Windows Vista SP 1.
- SSTP utilizes 2048-bit SSL/TLS certificates for authentication and 256-bit SSL keys for encryption.
- The biggest drawback to SSTP is that is basically a Microsoft-developed proprietary protocol and developers do not have access to the underlying code.

LIMITATIONS of a VPN

- There are more ways in which online entities can determine your identity which a VPN doesn't protect you from. When using Google Maps, for instance, you're often required to turn on GPS. This means Google Maps can see exactly where you are.
- Could result in slower internet


VPN's DURING THE COVID-19 PANDEMIC
Seen many people working from home, because home networks are less secure than office networks, are prone to attacks.
The most effective way to secure their communication channels between their workplaces and home is by routing through a VPN.
According to new investigations from Top10VPN, demand for VPNs increased by 44% over the second half of March and remains 22% higher than pre-pandemic levels.